

Chapter 14

L'Hospital's Rule

§ Section: 4.4

14.1 L'Hospital's Rule

L'Hospital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty.$$

In other word, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

Remark:

1. L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives, provided that the given conditions are satisfied. **It is especially important to verify the conditions regarding the limits of f and g before using l'Hospital's Rule.**
2. L'Hospital's Rule is also valid for **one-sided limits and for limits at infinity or negative infinity**; that is, “ $x \rightarrow a$ ” can be replaced by any of the symbols $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, or $x \rightarrow -\infty$.

3. If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ does not exist, it does NOT imply that $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ doesn't exist!

Example 1.1

Compute $\lim_{x \rightarrow \infty} \frac{\ln x + \cos x}{\ln x + \sin x}$.

$$\lim_{x \rightarrow \infty} \frac{f'}{g'} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \sin x}{\frac{1}{x} + \cos x} \quad \text{DNE}$$

$$\lim_{x \rightarrow \infty} \frac{f}{g} = \lim_{x \rightarrow \infty} \frac{1 + \frac{\cos x}{\ln x}}{1 + \frac{\sin x}{\ln x}} = 1. \quad \left(\frac{-1}{\ln x} \leq \frac{\cos x}{\ln x} \leq \frac{1}{\ln x} \right)$$

Example 1.2

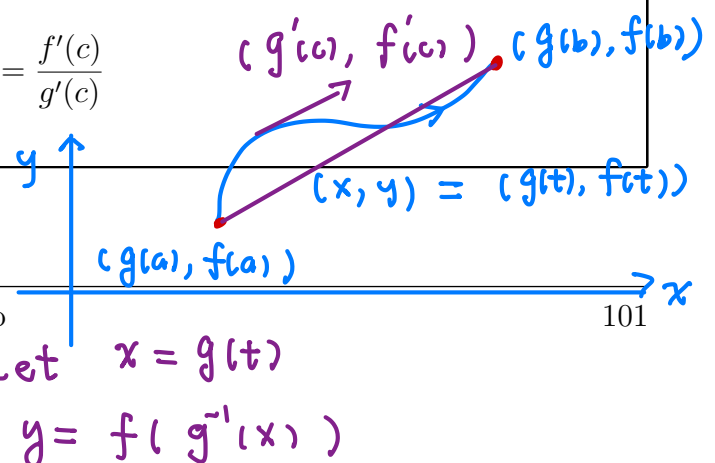
Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ and $g(x) = \sin x$. Compute $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$.

To prove l'Hospital's Rule, we need the following theorem.

Cauchy's Mean Value Theorem:

If $f(x), g(x)$ are continuous on $[a, b]$ and differentiable on (a, b) with $g'(x) \neq 0$ for all $x \in (a, b)$, then there is some $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$



Example 1.3

Prove Cauchy's Mean Value Theorem.

Consider $h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(x) - g(a))$. $h(x)$ is conti on $[a, b]$ and diff on (a, b) .

$$h(a) = f(a) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(a) - g(a)) = 0$$

$$h(b) = f(b) - f(a) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(b) - g(a)) = 0$$

Hence by Rolle's Thm, there is some $c \in (a, b)$ s.t.

$$h'(c) = 0. \text{ Therefore, } h'(c) = f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(c) = 0$$

Example 1.4

Prove l'Hospital's Rule.

$$\text{which means } \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Suppose that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$. Assume $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$. Let $F(x) = \begin{cases} f(x) & , \text{ for } x \in I \setminus \{a\} \\ 0 & , \text{ for } x = a \end{cases}$,

$$G(x) = \begin{cases} g(x) & , \text{ for } x \in I \setminus \{a\} \\ 0 & , \text{ for } x = a \end{cases}. \text{ For } x > a, F, G \text{ are conti on } [a, x] \text{ and diff on } (a, x).$$

Hence by Cauchy's MVT,

$$\frac{f(x)}{g(x)} = \frac{F(x)}{G(x)} = \frac{F(x) - F(a)}{G(x) - G(a)} \stackrel{\downarrow}{=} \frac{F'(c_x)}{G'(c_x)} = \frac{f'(c_x)}{g'(c_x)} \text{ for some } c_x \in (a, x)$$

$$\text{As } x \rightarrow a^+, \quad c_x \rightarrow a^+ \quad \text{and} \quad \frac{f'(c_x)}{g'(c_x)} \rightarrow L$$

$$(\because \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L) \text{ . Hence } \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Similarly, we can show that $\lim_{x \rightarrow a^-} \frac{f(x)}{g(x)} = L$.

$$\text{Thus, } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

14.2 Indeterminate Forms

Indeterminate Forms

1. Type $\frac{0}{0}$, Type $\frac{\infty}{\infty}$
2. Indeterminate Product: Type $0 \cdot \infty$
3. Indeterminate Difference: Type $\infty - \infty$
4. Indeterminate Powers: Type 0^0 ∞^0 1^∞

Type $\frac{0}{0}$, $\frac{\infty}{\infty}$

Example 2.1

$$\begin{aligned} \text{Find } \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}. & \quad \frac{\frac{0}{0}}{\frac{0}{0}} \quad \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} \quad \frac{\frac{0}{0}}{\frac{0}{0}} \quad \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{6x} \\ & = \lim_{x \rightarrow 0} \frac{\sec^2 x \cdot \tan x}{3x} = \lim_{x \rightarrow 0} \left(\frac{\sec^3 x}{3} \right) \left(\frac{\sin x}{x} \right) = \frac{1}{3} \end{aligned}$$

Example 2.2

$$\begin{aligned} \text{Find } \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}}}{x}. & \quad \frac{\frac{0}{0}}{\frac{0}{0}} \quad \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}} (-\frac{1}{x^2})}{1} = \lim_{x \rightarrow 0^-} -\frac{e^{\frac{1}{x}}}{x^2} \quad \frac{\frac{0}{0}}{\frac{0}{0}} \quad \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}} \frac{1}{x^3}}{2x} = \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}}}{2x^3} \\ & \quad \lim_{x \rightarrow 0^-} \frac{\frac{1}{x}}{e^{-\frac{1}{x}}} \quad \text{let } y = \frac{1}{x} \quad \lim_{y \rightarrow -\infty} \frac{y}{e^{-y}} \quad \frac{\frac{\infty}{\infty}}{\frac{\infty}{\infty}} \quad \lim_{y \rightarrow -\infty} \frac{1}{-e^{-y}} = 0 \end{aligned}$$

Example 2.3

Find $\lim_{x \rightarrow \infty} \frac{e^x}{x^n}$ for some $n \in \mathbb{N}$.

$$\begin{array}{c} \text{L'H} \parallel \frac{\infty}{\infty} \\ \lim_{x \rightarrow \infty} \frac{e^x}{n x^{n-1}} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{e^x}{n \cdot (n-1) x^{n-2}} = \dots = \lim_{x \rightarrow \infty} \frac{e^x}{n!} = \infty \end{array}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty \quad \Rightarrow \quad e^x \gg x^n \text{ for } x \text{ large.}$$

Example 2.4

Find $\lim_{x \rightarrow \infty} \frac{\ln x}{x^a}$ for some $a > 0$.

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^a} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{a \cdot x^{a-1}} = \lim_{x \rightarrow \infty} \frac{1}{a x^a} = 0.$$

$$\ln x \ll x^a \text{ for } x \text{ large.}$$

$$\text{e.g. } \ln x \ll \sqrt{x}, \quad \ln x \ll \sqrt[3]{x}, \quad \ln x \ll \sqrt[n]{x}, \quad n \in \mathbb{N}.$$

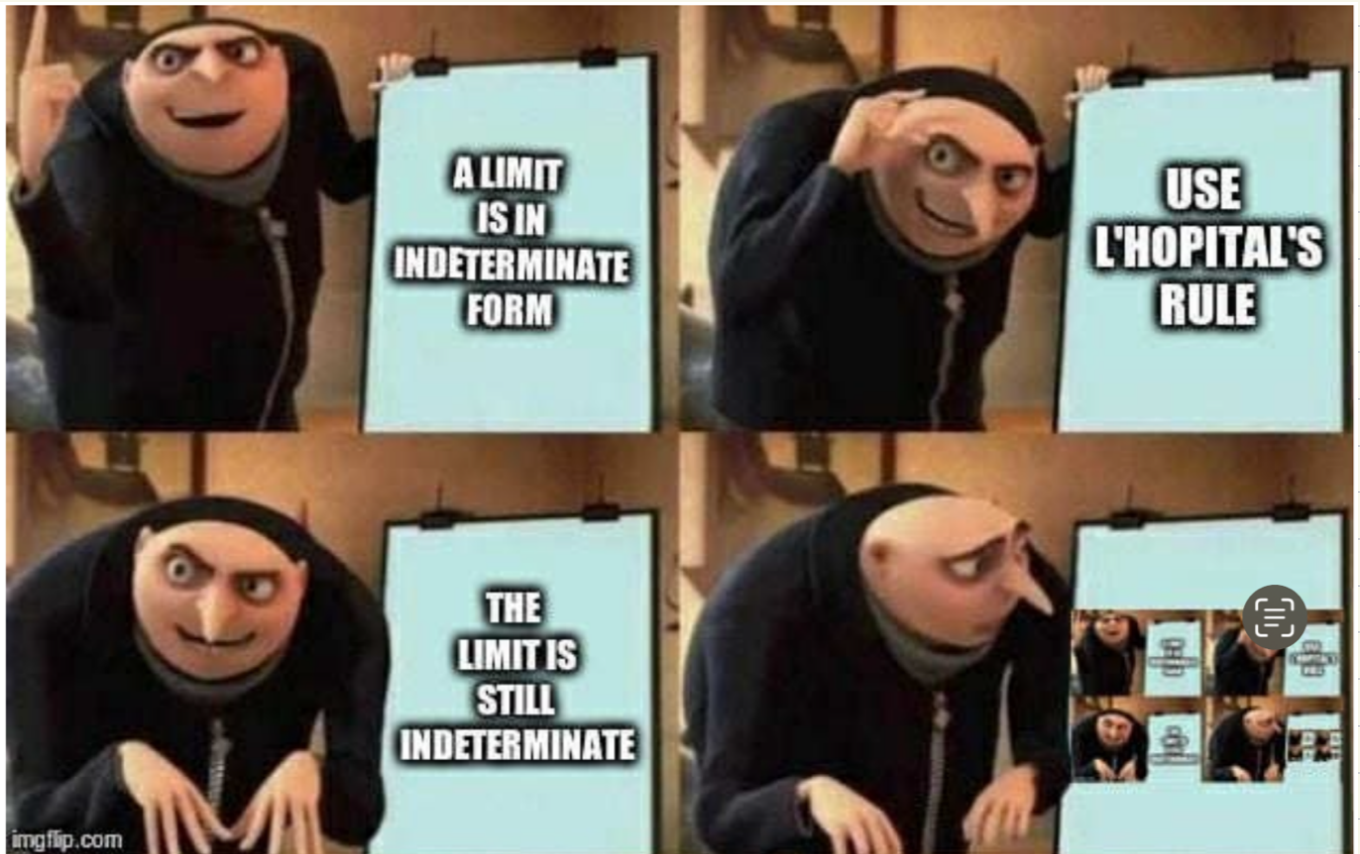
Example 2.5

Suppose that f is differentiable and $f''(a)$ exists. Find $\lim_{x \rightarrow a} \frac{f(x) - [f(a) + f'(a)(x-a)]}{(x-a)^2}$.

$$\begin{aligned} & \stackrel{\frac{0}{0}}{\text{L'H}} \lim_{x \rightarrow a} \frac{f'(x) - f'(a)}{2(x-a)} \\ &= \frac{1}{2} f''(a) \end{aligned}$$

In fact, by Cauchy's Mean Value Theorem,

$$f(x) - L(x) = \frac{1}{2} f''(c)(x-a)^2 \quad \text{for some } c \text{ between } a \text{ and } x.$$



Type $0 \cdot \infty$ **Example 2.6**

$$\text{Find } \lim_{x \rightarrow 0^+} x \ln x. = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \stackrel{\frac{0}{\infty}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0^+} (-x) = 0$$

$$\text{Ex: } \lim_{x \rightarrow 0^+} x^a \ln x = 0 \quad \text{for all } a > 0.$$

Example 2.7

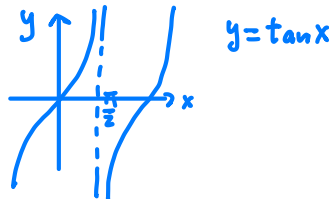
$$\lim_{x \rightarrow 0^+} \sin(\sqrt{x}) \ln x.$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\sin(\sqrt{x})}{\sqrt{x}} \right) (\sqrt{x} \cdot \ln x) = 1 \cdot 0 = 0.$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{\sqrt{x}}} \stackrel{\frac{0}{\infty}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{2} \frac{1}{x^{3/2}}} = \lim_{x \rightarrow 0^+} -2\sqrt{x} = 0.$$

Example 2.8

$$\text{Find } \lim_{x \rightarrow 1^+} \ln x \cdot \tan\left(\frac{\pi}{2}x\right).$$



$$= \lim_{x \rightarrow 1^+} \frac{\ln x}{\cot\left(\frac{\pi}{2}x\right)}$$

$$\stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 1^+} \frac{\frac{1}{x}}{-\csc^2\left(\frac{\pi}{2}x\right) \cdot \frac{\pi}{2}} = -\frac{2}{\pi}.$$

Example 2.9

$$\text{Find } \lim_{x \rightarrow -\infty} x(e^{\frac{1}{x}} - 1). = \lim_{x \rightarrow -\infty} \frac{(e^{\frac{1}{x}} - 1)}{\frac{1}{x}} \stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow -\infty} \frac{e^{\frac{1}{x}} \cdot \cancel{(-\frac{1}{x^2})}}{\cancel{(-\frac{1}{x^2})}} = 1.$$

Type $\infty - \infty$

Example 2.10

$$\text{Find } \lim_{x \rightarrow (\frac{\pi}{2})^-} \sec x - \tan x. = \lim_{x \rightarrow (\frac{\pi}{2})^-} \frac{1 - \sin x}{\cos x} \stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow (\frac{\pi}{2})^-} \frac{-\cos x}{-\sin x} = 0$$

Example 2.11

$$\text{Find } \lim_{x \rightarrow \infty} x - \ln x. = \lim_{x \rightarrow \infty} \ln e^x - \ln x = \lim_{x \rightarrow \infty} \ln \frac{e^x}{x}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{e^x}{x} \stackrel{\frac{\infty}{\infty}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow \infty} \frac{e^x}{1} = \infty$$

$$\therefore \lim_{x \rightarrow \infty} \ln \left(\frac{e^x}{x} \right) = \infty.$$

Example 2.12

$$\text{Find } \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right). = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x(e^x - 1)}$$

$$\stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{e^x - 1}{e^x - 1 + x \cdot e^x} \stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{e^x}{e^x + e^x + x \cdot e^x} = \frac{1}{2}.$$

Example 2.13

Find $\lim_{x \rightarrow \infty} \left(x - x^2 \sin \frac{1}{x} \right) = \lim_{x \rightarrow \infty} x \left(1 - x \sin \frac{1}{x} \right)$

$$= \lim_{x \rightarrow \infty} \frac{1 - x \sin(\frac{1}{x})}{\frac{1}{x}} \quad \text{let } y = \frac{1}{x} \quad \lim_{y \rightarrow 0} \frac{1 - \frac{\sin y}{y}}{y}$$

$$= \lim_{y \rightarrow 0} \frac{y - \sin y}{y^2} \quad \frac{0}{0} \quad \lim_{y \rightarrow 0} \frac{1 - \cos y}{2y} \quad \frac{0}{0} \quad \lim_{y \rightarrow 0} \frac{\sin y}{2} = 0.$$

Indeterminate Powers 0^0 , 1^∞ , ∞^0

$\lim_{x \rightarrow a} f(x)^{g(x)}$ is an **indeterminate power** if and only if $\lim_{x \rightarrow a} g(x) \ln(f(x))$ is an **indeterminate product**.

$f(x)^{g(x)}$	$g(x) \ln(f(x))$
0^0	$0 \cdot \infty \quad \because \lim_{x \rightarrow a} g(x) = 0, \lim_{x \rightarrow a} \ln f(x) = -\infty.$
1^∞	$\infty \cdot 0 \quad \because \lim_{x \rightarrow a} g(x) = \infty, \lim_{x \rightarrow a} \ln f(x) = 0.$
∞^0	$0 \cdot \infty \quad \because \lim_{x \rightarrow a} g(x) = 0, \lim_{x \rightarrow a} \ln f(x) = \infty.$

Example 2.14

Find $\lim_{x \rightarrow 0^+} x^x$.

$$\text{sol: } \ln(x^x) = x \cdot \ln x$$

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \quad \frac{0}{0} \quad \lim_{x \rightarrow 0^+} \frac{-\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0^+} (-x) = 0$$

Hence $\lim_{x \rightarrow 0^+} x^x = e^0 = 1.$

Exercise 2.1

Let $f(x) = |x|^x$ for $x \neq 0$, $f(0) = 1$. Is $f(x)$ differentiable at $x = 0$?

Example 2.15Find $\lim_{x \rightarrow 0^+} x^{x^x}$.

$$\therefore \lim_{x \rightarrow 0^+} x^x = 1$$

$$\therefore \lim_{x \rightarrow 0^+} x^{(x^x)} = \lim_{x \rightarrow 0^+} 1^1 = 1.$$

Example 2.16Find $\lim_{x \rightarrow 0} \left(\frac{2^x + 3^x}{2}\right)^{\frac{1}{x}}$, $\lim_{x \rightarrow \infty} \left(\frac{2^x + 3^x}{2}\right)^{\frac{1}{x}}$.

$$\text{sol: } \ln \left[\left(\frac{2^x + 3^x}{2} \right)^{\frac{1}{x}} \right] = \frac{1}{x} \ln \left(\frac{2^x + 3^x}{2} \right) = \frac{\ln(2^x + 3^x) - \ln 2}{x}$$

$$\lim_{x \rightarrow 0} \frac{\ln(2^x + 3^x) - \ln 2}{x} \stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{\frac{\ln 2 \cdot 2^x + \ln 3 \cdot 3^x}{2^x + 3^x}}{1} = \frac{\ln 2 + \ln 3}{2}$$

$$\text{Hence } \lim_{x \rightarrow 0} \left(\frac{2^x + 3^x}{2} \right)^{\frac{1}{x}} = e^{\frac{\ln 2 + \ln 3}{2}} = \sqrt{e^{\ln 2 + \ln 3}} = \sqrt{6}$$

Example 2.17Find $\lim_{x \rightarrow 0^+} (1 + \sin 2x)^{\cot 3x}$.

$$\text{sol: } \ln \left[(1 + \sin 2x)^{\cot 3x} \right] = \cot 3x \cdot \ln(1 + \sin 2x) = \frac{\ln(1 + \sin 2x)}{\tan 3x}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 2x)}{\tan 3x} \stackrel{\frac{0}{0}}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0^+} \frac{\frac{2 \cos 2x}{1 + \sin 2x}}{3 \sec^2 3x} = \frac{2}{3}$$

$$\text{Hence } \lim_{x \rightarrow 0^+} (1 + \sin 2x)^{\cot 3x} = e^{2/3}.$$

Example 2.18Find $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$ and $\lim_{x \rightarrow \infty} (1+x)^{\frac{1}{x}}$.

$$\begin{aligned} \ln(1+x)^{\frac{1}{x}} &= \frac{1}{x} \cdot \ln(1+x) = \frac{\ln(1+x)}{x} \longrightarrow \begin{cases} 1, & \text{as } x \rightarrow 0 \\ 0, & \text{as } x \rightarrow \infty \end{cases} \\ \Rightarrow \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} &= e^1 = e. \\ \lim_{x \rightarrow \infty} (1+x)^{\frac{1}{x}} &= e^0 = 1 \end{aligned}$$

Example 2.19Find $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{\sin(x^2)}} = e^{-\frac{1}{6}}$.

$$\text{Sol: } \ln \left[\left(\frac{\sin x}{x} \right)^{\frac{1}{\sin(x^2)}} \right] = \frac{1}{\sin(x^2)} \cdot \left[\ln \left(\frac{\sin x}{x} \right) \right]$$

$$= \frac{\ln|\sin x| - \ln|x|}{\sin(x^2)}$$

$$\lim_{x \rightarrow 0} \frac{\ln|\sin x| - \ln|x|}{\sin(x^2)} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\frac{\cos x}{\sin x} - \frac{1}{x}}{2x \cdot \cos(x^2)}$$

$$= \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{2x^2 \sin x \cdot \cos(x^2)} = \lim_{x \rightarrow 0} \left(\frac{x \cos x - \sin x}{2x^2 \sin x} \right) \cdot \left(\frac{1}{\cos(x^2)} \right) = -\frac{1}{6}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{2x^2 \sin x} &\stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\cancel{\cos x} - x \cancel{\sin x} - \cancel{\cos x}}{4x \sin x + 2x^2 \cos x} = \lim_{x \rightarrow 0} \frac{-\frac{\sin x}{x}}{4 \frac{\sin x}{x} + 2 \cos x} \\ &= -\frac{1}{6} \end{aligned}$$